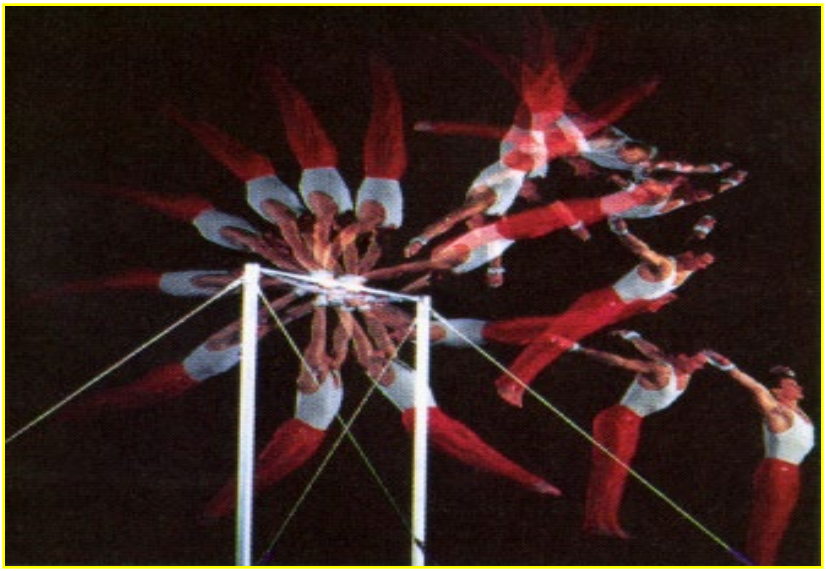
PHYC10009 Foundations of Physics – Lecture 2



Professor Martin Sevior

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Physics, Room 501

(Please put your mobile phone in silent mode.)

Staff Student Liaison Committee Representative (SSLC).

Survey class, summarise to SSLC

Provide feedback to class.

Two meetings during semester

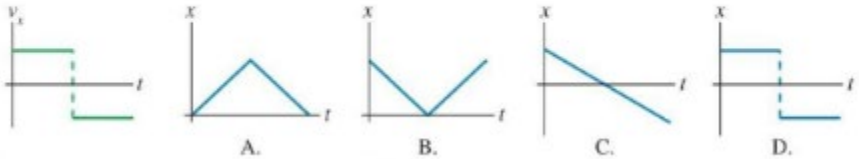
- Late in semester, includes fine dining at University House

Good for your CV!

If you would like to nominate to be SSLC rep, please send me an email.

We will hold our election next Tuesday.

Which position-time graph (A-D) corresponds to the given velocity-time graph?



A

B

C

D

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Constant acceleration - G2.5

$$v_f = v_i + a\Delta t \qquad (1)$$

$$\Delta x = v_i \Delta t + \frac{1}{2} a(\Delta t)^2 \qquad (2)$$

From (1): $a\Delta t = v_f - v_i$

Sub into (2): $\Delta x = v_i \Delta t + \frac{1}{2}\Delta t(v_f - v_i)$

$$\Delta x = \frac{1}{2}(v_i + v_f)\Delta t \qquad (3)$$

Constant acceleration - G2.5

$$v_f = v_i + a\Delta t \qquad (1)$$ ← No Δx

$$\Delta x = v_i \Delta t + \frac{1}{2} a (\Delta t)^2 \qquad (2)$$ ← No v_f

$$\Delta x = \frac{1}{2}(v_i + v_f)\Delta t \qquad (3)$$ ← No a

From (1): $\Delta t = \dfrac{v_f - v_i}{a}$

Sub into (3): $\Delta x = \dfrac{1}{2}(v_f + v_i)\dfrac{(v_f - v_i)}{a} = \dfrac{v_f^{\,2} - v_i^{\,2}}{2a}$

$$2a\Delta x = v_f^{\,2} - v_i^{\,2} \qquad (4)$$ ← No Δt

Constant Acceleration summary

$$v_f = v_i + a\Delta t \qquad (1)$$

$$\Delta x = v_i \Delta t + \frac{1}{2} a (\Delta t)^2 \qquad (2)$$

$$\Delta x = \frac{1}{2}(v_i + v_f)\Delta t \qquad (3)$$

$$2a\Delta x = v_f^{\,2} - v_i^{\,2} \qquad (4)$$

Example problem

Channel Nine News ran a story about the dangers of bunk beds (17 Feb 2009) which included the following statistic:

> "Government statistics show at least 3850 children under 15 injure themselves on bunk beds every year."

Assume: Height of bunk bed h = 1.6m, mass of child m = 45kg; child hits floor with body approximately horizontal

Write an expression for the velocity of a child as it strikes the floor, assuming the child rolls out of the bed (i.e. falls height h from a starting velocity of zero in the vertical direction) and calculate that velocity.

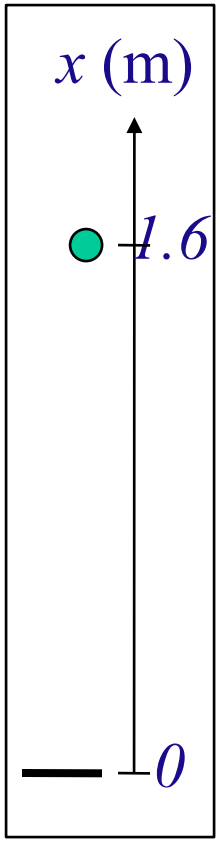
Example

$$v_f = v_i + a\Delta t$$
$$\Delta x = v_i \Delta t + \frac{1}{2}a(\Delta t)^2$$
$$\Delta x = \frac{1}{2}(v_i + v_f)\Delta t$$
$$2a\Delta x = v_f^{\,2} - v_i^{\,2}$$

a = g

We know Δx, v_i, a, so...

$$1.6 = 0 \times \Delta t + \frac{1}{2}a(\Delta t)^2$$

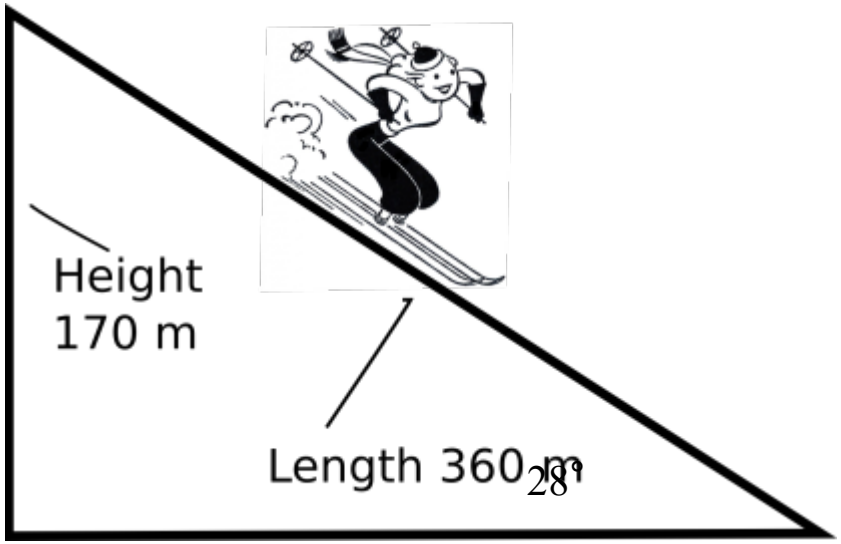


$$\Delta t = \sqrt{\frac{3.2}{g}} = \sqrt{\frac{3.2}{9.8}} = 0.57s$$

Can now calculate v_f in several ways. For example…

$$v_f = 0 + (9.8)(0.57) = 5.59ms^{-1}$$

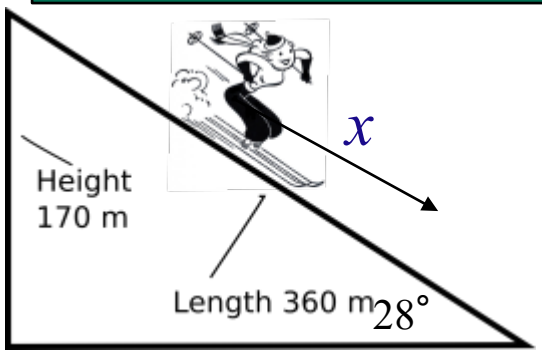
Example – The skier



In a speed-skiing event, a skier starts from rest and travels down a slope of length 360 m with a constant acceleration of 4.63 ms^{-2}.

- What is the skier's speed at the end of the slope?
- How long does it take the skier to reach the bottom?

Example – the skier



$$v_f = v_i + a\Delta t$$
$$\Delta x = v_i \Delta t + \frac{1}{2}a(\Delta t)^2$$
$$\Delta x = \frac{1}{2}(v_i + v_f)\Delta t$$
$$2a\Delta x = v_f^{\,2} - v_i^{\,2}$$

$$\frac{57.7\,\cancel{\text{m}}}{\cancel{\text{s}}} \times \frac{1\text{ km}}{1000\,\cancel{\text{m}}} \times \frac{60\,\cancel{\text{min}}}{\text{hr}} \times \frac{60\,\cancel{\text{s}}}{\cancel{\text{min}}}$$
$$= 208\text{ km hr}^{-1}$$ *(Fast!)*

- $\Delta x = 360$ m, $a = 4.63$ ms^{-2}.
- What is the skier's speed at the end of the slope?
  - We need to find v_f.
  - Don't know Δt, so use the equation which does not involve Δt.

$$v_f^{\,2} = v_i^{\,2} + 2a\Delta x$$
$$v_f = \sqrt{0^2 + 2(4.63)(360)} = 57.7\text{ ms}^{-1}$$

- How long does it take the skier to reach the bottom?
  - Need to find Δt.
  - Since we now know all the other quantities, can use any equation involving Δt.

$$\Delta x = \frac{1}{2}a(\Delta t)^2 \quad \text{so} \quad \Delta t = \sqrt{\frac{2\Delta x}{a}} = \sqrt{\frac{2(360)}{4.63}} = 12.5\text{ s}$$

Is there a better way to calculate Δt using a different equation? Exercise!

GAMSAT Style Question

In a human body, the speed of blood through the arterial pathway generally decreases with distance from the heart.

In an experiment it was observed that the speed of blood flow was 0.30 m s −1 at one point in an artery and 0.20 m s −1 at another point 0.50 m further along the arterial pathway. Assume that the blood flow slowed at a constant rate.

What was the deceleration of the blood between the two points?

A	0.01 m s −2
B	0.05 m s −2
C	0.10 m s −2
D	0.20 m s −2

$$v_f = v_i + a\Delta t$$

$$\Delta x = v_i \Delta t + \frac{1}{2} a(\Delta t)^2$$

$$\Delta x = \frac{1}{2}(v_i + v_f)\Delta t$$

$$2a\Delta x = v_f^{\ 2} - v_i^{\ 2}$$

Use constant acceleration formula:

$$2a\Delta x = v_f^{\ 2} - v_i^{\ 2} \qquad (4)$$

$$a = \frac{v_f^{\ 2} - v^2{}_i}{2\Delta x} = \frac{0.2^2 - 0.3^2}{2 \times 0.5} = -0.05 \text{ ms}^{-2}$$

Vectors – Key points

Vector possesses

- Magnitude
- Direction

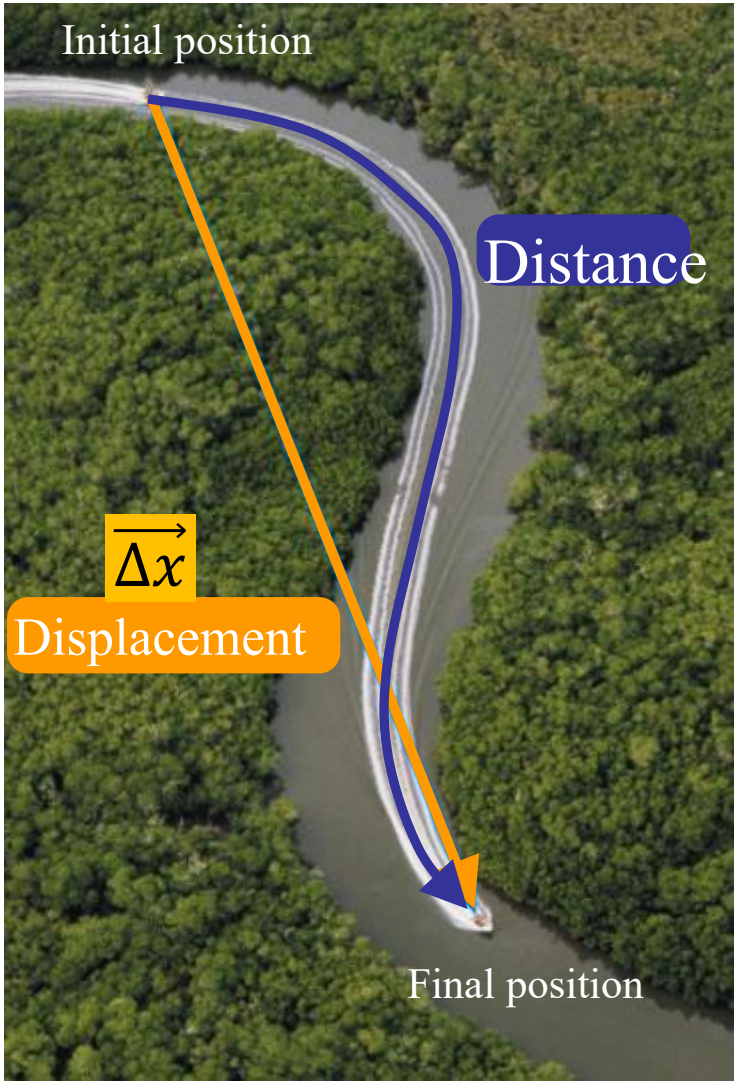
Vectors can be manipulated

- Using components
- Graphically

Unit vectors

Addition and subtraction can be performed component by component

Vectors: displacement vs distance G3.1



Displacement depends *only* on initial and final positions

Distance depends on the exact path taken

Displacement has a fixed unique direction (from initial to final point)

Distance has no direction – it is just the length of the actual path taken

i.e. Distance is <u>not</u> a vector
Displacement is a vector

Scalars and vectors - G 3.1

Scalars

Described by a single number with a unit

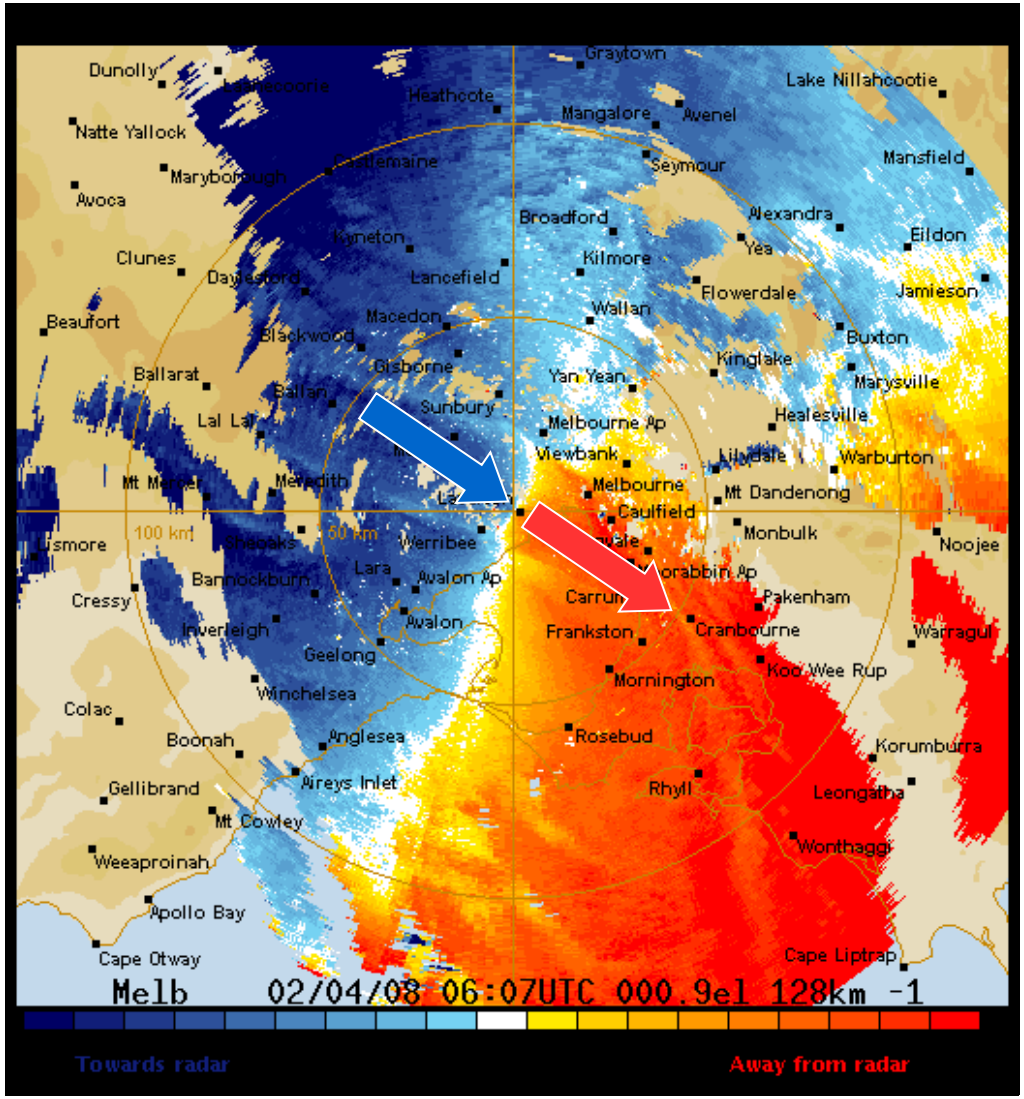
eg time, temperature, speed, distance

Vectors

Directional aspect

Magnitude *and* direction

eg wind velocity, displacement



Properties of vectors G 3.1

Need to specify *two* pieces of information

Magnitude including relevant units

Direction

Represent by an arrow

Magnitude = Length of arrow

Direction = Direction of arrow

Wind example: *"What is the wind velocity?"*

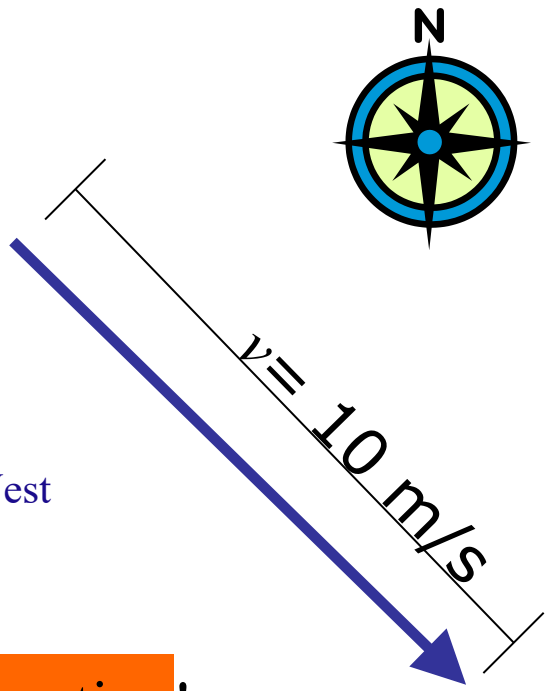
Wind speed ~ 10 m/s

Direction – From the North West

ANSWER: The velocity is 10 m/s from the North West

Units!

Direction!



1.5 Vectors and scalars

Scalars

E.g. mass, volume, distance, speed

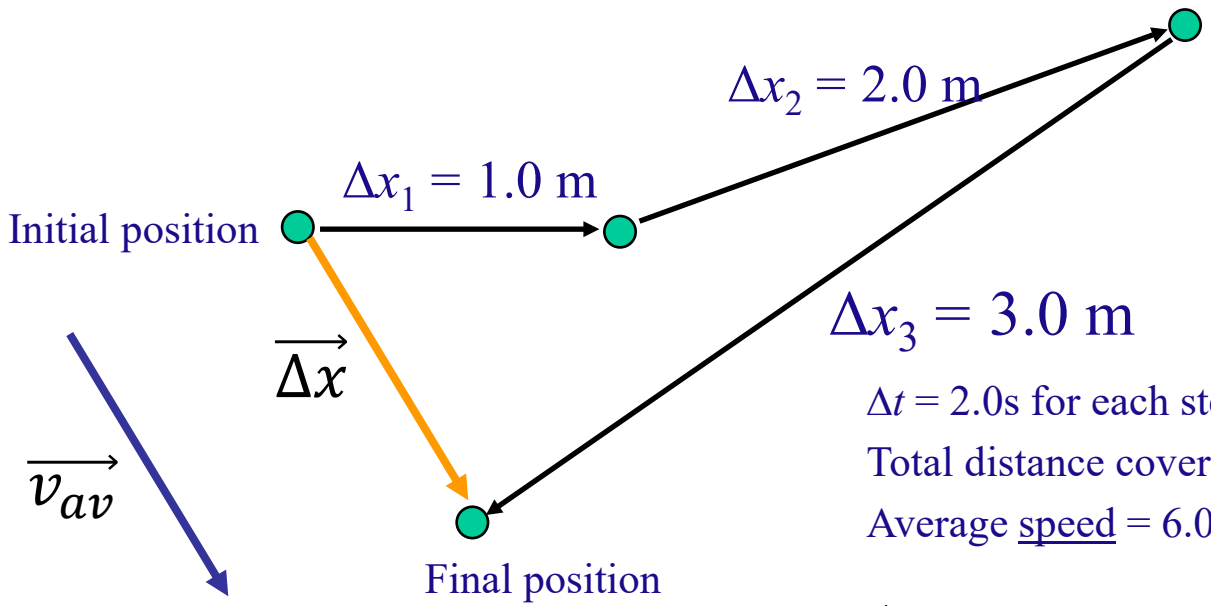
Magnitude only

Vectors

E.g. displacement, velocity, acceleration

Magnitude and *direction*

$\vec{v}$ **v** (bold face) $v = |\vec{v}|$



$\Delta x_3 = 3.0$ m

$\Delta t = 2.0$s for each step

Total distance covered = 6.0 m.

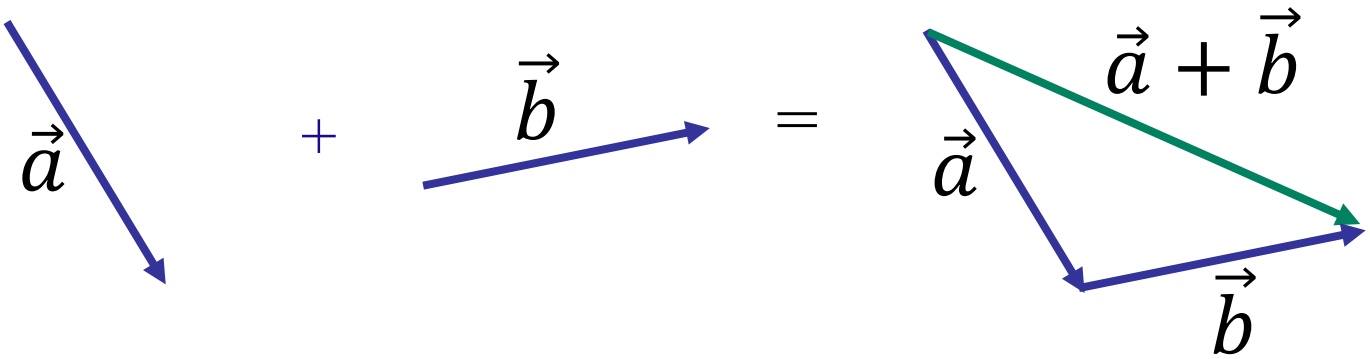
Average <u>speed</u> = 6.0 m / 6.0 s = 1.0 m/s.

$$|\overrightarrow{v_{av}}| = v_{av} = \frac{1.5m}{6.0s} = 0.25\ m/s$$ in direction as shown

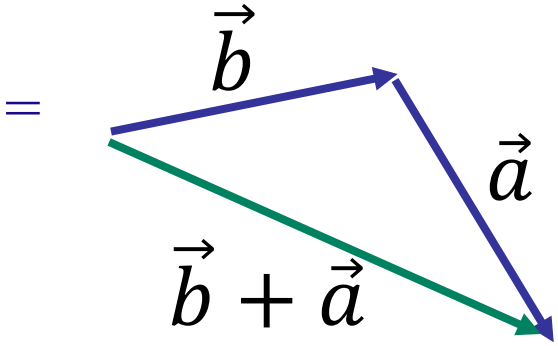
Adding vectors G.2

Place vectors head-to-tail (order doesn't matter)

Result is vector pointing from tail of first to head of second

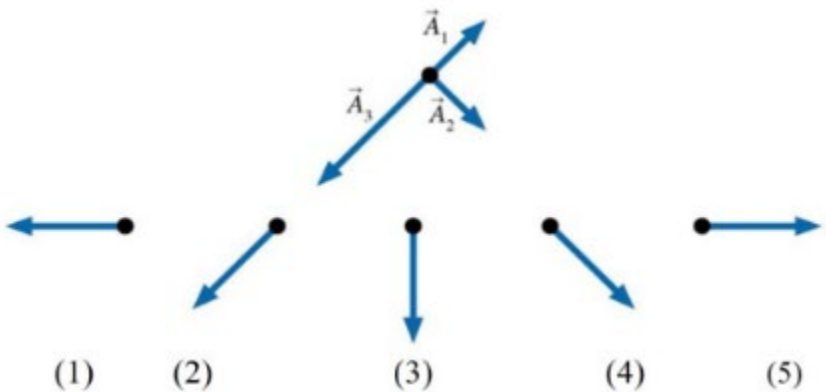


Note: a vector *only* has magnitude and direction. It is not 'fixed' at a particular point in space. It can be moved around, as long as its magnitude and direction do not change.



Vector Addition - 1

Which figure shows $\vec{A}_1 + \vec{A}_2 + \vec{A}_3$?



(1)

(2)

(3)

(4)

(5)

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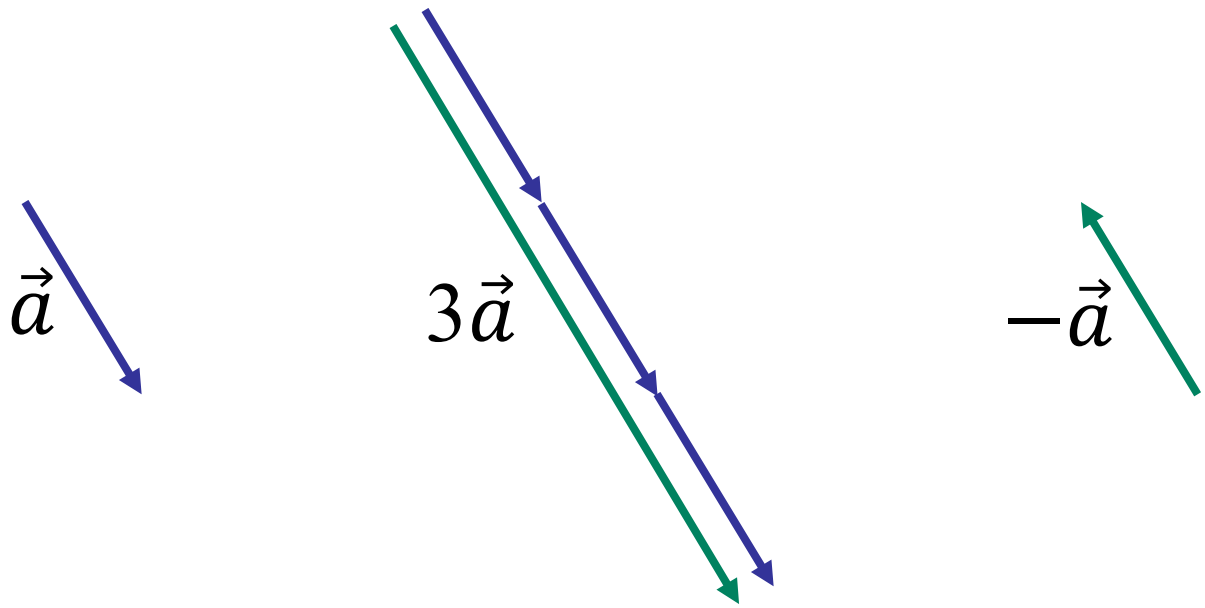
Multiplying a vector by a scalar

Multiply *magnitude* of vector by scalar

Direction unchanged

Multiplication by a negative number reverses direction of vector

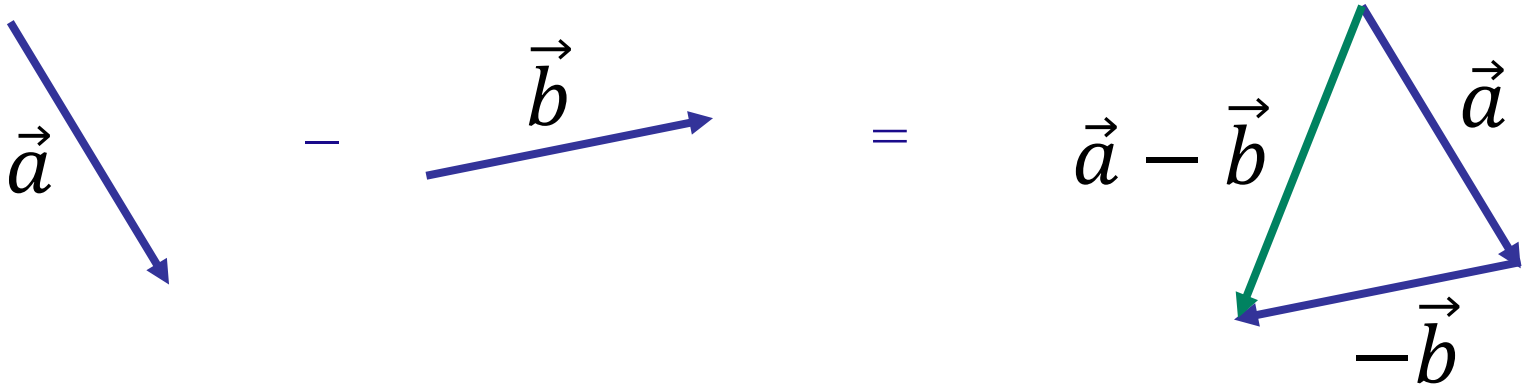


Subtracting vectors - G3.3

Turn second vector around (multiply by –1) and add

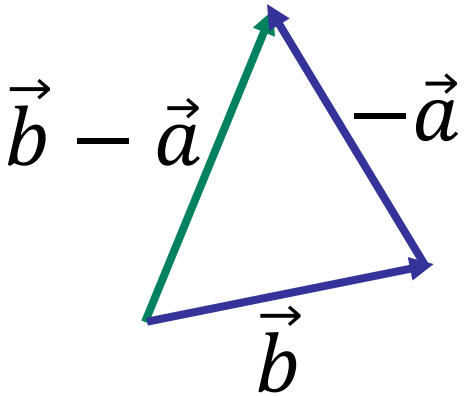
Take care! Don't reverse direction of wrong vector

Order matters, as you can see here:

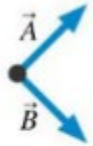


$$\boxed{\vec{a} - \vec{b} \equiv \vec{a} + (-\vec{b})}$$

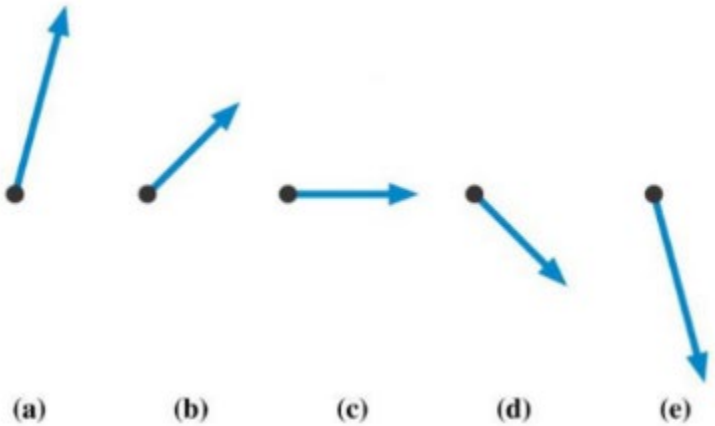
$$\vec{b} - \vec{a} \quad = -(\vec{a} - \vec{b})$$



Vector Addition - 2

Given 

which figure shows $2\vec{A} - \vec{B}$?



(a)

(b)

(c)

(d)

(e)

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Summary

Constant Acceleration

$$v_f = v_i + a\Delta t \qquad (1)$$

$$\Delta x = v_i \Delta t + \frac{1}{2} a (\Delta t)^2 \qquad (2)$$

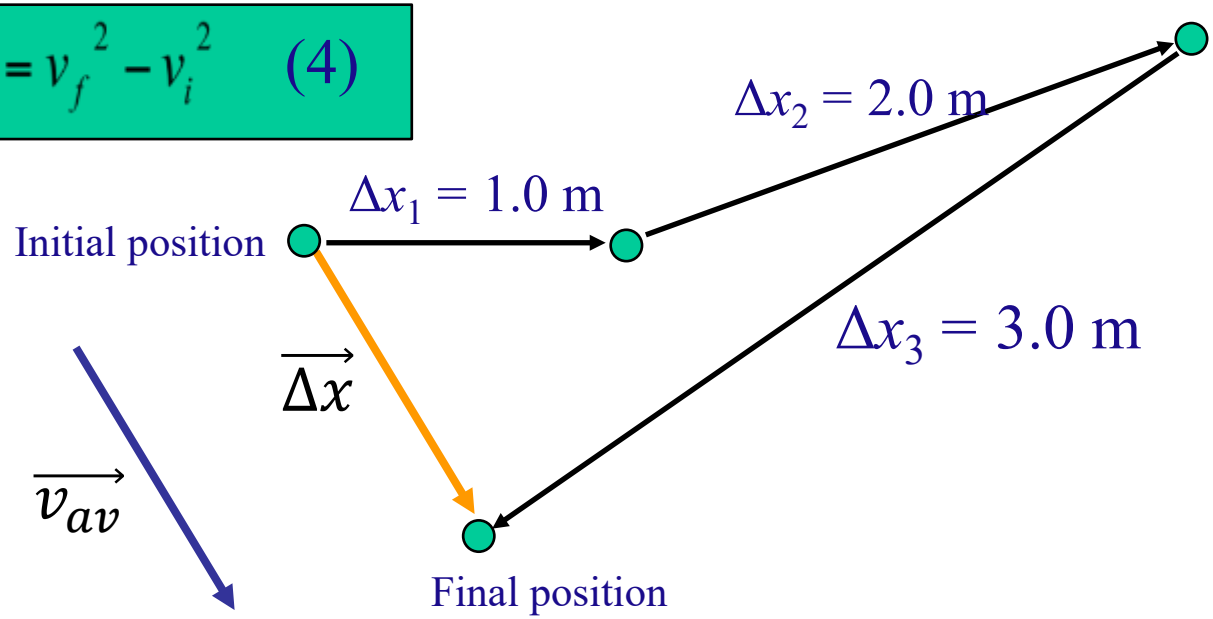
$$\Delta x = \frac{1}{2}(v_i + v_f)\Delta t \qquad (3)$$

$$2a\Delta x = v_f^{\,2} - v_i^{\,2} \qquad (4)$$

Vectors

Need to specify *two* pieces of information

- Magnitude including relevant units
- Direction

Represent by an arrow

- Magnitude = Length of arrow
- Direction = Direction of arrow